

Adopting an Ecological Approach to Misinformation: Understanding the Broader Impacts on Online Communities

Zhila Aghajari
zha219@lehigh.edu
Lehigh University
Bethlehem, PA, USA

ABSTRACT

Prior work has acknowledged the significance of social and community oriented factors in the spread and impacts of misinformation. However, interventions have largely focused on individual pieces of false and misleading content as misinformation, de-emphasizing the role of community-oriented factors that are involved in and contribute to the broad impacts of misinformation. My dissertation highlights the consequences that arise from such an individualistic focus. To account for the broader scope of this misinformation and its impacts, it proposes adopting an ecological perspective to misinformation. Employing this perspective, my work examines the community-level impacts of misinformation, from shaping perceptions about online communities to shifting the way online communities interpret and respond to the world's events. Indeed, understanding and accounting for such significant impacts of misinformation on online communities is important if we hope to address the broader, systemic nature of misinformation and its effects at the community level.

CCS CONCEPTS

• **Human-centered computing** → **Empirical studies in HCI**;
Empirical studies in collaborative and social computing.

KEYWORDS

Misinformation, Online Communities, Community-level Impacts, Perceived Social Norms, Framing, Meaning Constructions

ACM Reference Format:

Zhila Aghajari. 2023. Adopting an Ecological Approach to Misinformation: Understanding the Broader Impacts on Online Communities. In *Computer Supported Cooperative Work and Social Computing (CSCW '23 Companion)*, October 14–18, 2023, Minneapolis, MN, USA. ACM, New York, NY, USA, 4 pages. <https://doi.org/10.1145/3584931.3608915>

1 INTRODUCTION

Misinformation has significant consequences, including political, economic, and social ones. Examples include the effects of misinformation on the US 2016 and 2020 presidential elections [12, 26, 30], its impacts on India's economy during the 2019 coronavirus disease

outbreaks [4], and its impacts on attitudes toward COVID-19 vaccines [14, 25]. In several cases, online misinformation even inspired violent attacks in physical interactions [e.g., 2, 28].

Numerous interventions have been designed to address misinformation and its impacts (e.g., fact-checking, signaling credibility of the content, signaling credibility of the source of content, and providing more perspectives on the content) [e.g., 1, 3, 5, 10]. Despite the progress made in addressing misinformation, however, this phenomenon is clearly a persistent and critical challenge that requires continued attention [22, 33].

Part of the challenge involves conceptualizing what constitutes misinformation. Some researchers define misinformation as “false or misleading information” [22, 33]. Some others describe misinformation as “a claim that contradicts or distorts common understanding of verifiable facts” [19]. Both these definitions implicitly treat misinformation as individual pieces of information. While addressing pieces of false and misleading information is important, the full scope of the problem exceeds the bounds of such definitions. For example, misinformation can contribute to creating an information space in which individuals become skeptical of all claims, even the ones that are true, and lead to a breakdown in trust in information space, in science, and in authorities [23, 24].

Alternatively, we could conceive of misinformation as a broad phenomenon that transcends individual pieces of false or misleading content. In this view, misinformation is not an isolated phenomenon. Rather, it is situated in a complex information ecosystem where multiple factors interact and shape individuals' perceptions of the world's events, thereby impacting their responses to those events. The impacts of misinformation could similarly extend beyond misleading individuals about individual pieces of content. For example, the general distrust in science mentioned above arguably do not result from any single, individual piece of content. Instead, they also pertain to social factors surrounding such content, including (perceived) norms of a community, and how communities discuss, view, and interpret the world's events and the surrounding issues (i.e., framing) [16, 17].

Thus, to address misinformation and its broad impacts, it may be useful to incorporate approaches that extend beyond identifying and correcting individual pieces of untrue content. Put concisely, it may be beneficial to adopt **an ecological perspective on the phenomenon of misinformation** that considers the various elements within the information ecosystem that are involved in individuals' understanding of and their response to the world's events. Such a comprehensive perspective may enable us to offer different strategies for approaching the problem of misinformation and its extensive ramifications.

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

CSCW '23 Companion, October 14–18, 2023, Minneapolis, MN, USA

© 2023 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-0129-0/23/10.

<https://doi.org/10.1145/3584931.3608915>

A variety of methods can be useful in providing an ecological perspective, from social scientific experiments to computational analysis techniques. For example, examining the concept of framing [15, 17], which refers to the processes by which people interpret events around them and construct their understanding of those event, can provide a conceptual framework that allows us to adopt an ecological perspective when examining misinformation and its impacts.

Therefore, my work adopts an ecological approach to misinformation to understand the broader scope of this phenomenon and its impacts. To do so, it designs and conducts a series of studies and examines the following research questions.

RQ1: How do current interventions that are designed to address misinformation define and scope this phenomenon?

RQ2: What are the mechanisms by which misinformation affects community level perceptions, especially perceptions about community's norms, which lead to broader impacts on online communities?

RQ3: How can computational models be designed to provide an ecological account of framing processes, as evidenced in the language used in online communities? ¹

RQ4: What role does the prevalence of false and misleading content play in the processes by which online communities interpret and construct meaning about the world's events (i.e., framing)?

The first two research questions have already been examined using two complementary studies (§2). I am proposing two other studies to attend to the other two research questions stated above (§3).

2 WORK COMPLETED AND THE KEY FINDINGS

2.1 The Need to Expand Our Vision Beyond an Individualistic Focus to Misinformation

To explore the ways by which prior work defines and scopes the broad phenomenon of misinformation, my work conducts a critical literature review (Described in detail in Aghajari et al. [6]). Specifically, it examines how these approaches account for the diverse underlying mechanisms that are involved misinformation and its impacts.

Key findings show that previous studies maintains a primarily individualistic focus to misinformation. Specifically, they tend to consider misinformation as isolated pieces of false and misleading content that needs to be addressed. However, this review paper argues that such an individualistic focus de-emphasizes the role of social factors in the spread and impacts of misinformation, and draws attentions away from the systematic nature and consequences of misinformation at the community level. Thus, to account for the broader scope of misinformation and its community-level impacts, this review highlights the need to expand our vision beyond individual pieces of content and considering for the community-oriented factors that are involved in this phenomenon. To do so, my dissertation adopts an ecological perspective on the phenomenon of

misinformation that considers the various elements within the information ecosystem, in particular the community oriented factors, that are involved in and contribute to misinformation and its broad impacts.

2.2 Misinformation and its Broad Impacts on Community Level Perceptions

To account for the community oriented aspects of misinformation, the work proposed in this chapter of my dissertation examines the significant role of perceived norms. Indeed, perceived norms not only contribute to the impacts of misinformation at the community level, but also are influenced by misinformation themselves [13, 21, 27].

To examine the potential mechanisms by which misinformation impacts individuals' perceptions of a community's norms, as well as the way such perceptions may then lead to broader community level impacts, my work conducts two controlled, experimental studies (Described in detail in Aghajari et al. [7]). Key findings suggest that the prevalence of false or misleading content, even when constituting a minority of posts in a community, has significant impacts on perceptions about a community, both in terms of perceptions about a community's norms and about that community's broader beliefs. However, community response to false and misleading content can play a greater role and mitigate the effects of such content on the perceptions about a community and its norms. This study offers valuable implications about leveraging the role of community responses in the development of policies, design, and governance for online communities to address the broader consequences of misinformation at the community level.

3 WORK IN PROGRESS: MISINFORMATION AND ITS ROLE IN FRAMING PROCESSES

In line with the broader societal view of misinformation in my dissertation, this chapter of my work examines the role that misinformation play in the processes by which individuals and online communities interpret and construct their understanding of the world's events.

This focus aligns with the concept of *framing* ², as defined in sociological studies [9, 16]. Put concisely, in these studies, framing refers to the processes by which individuals construct their understandings of and respond to events around them.

Therefore, examining the processes involved in framing may provide a framework to explore how false and misleading content interplays with community responses, and how it influences individuals' interpretation of and their response to the world's events. In this way, framing allows us to move beyond considering misinformation as individual pieces of false and misleading content, and to study the broader scope of misinformation and its impacts on how people view the world's events and respond to them.

To be able to account for the broad, heterogeneous contexts in which framing processes occur, this work will examine different

¹This research question is stated to help design computational techniques that allow to study processes involved in framing, and to enable my dissertation to then examine the ways misinformation may play a role in these processes.

²While concept of framing is studied in different fields – from political communication [e.g., 29], to psychology [e.g., 31], to sociology [e.g., 18], to behavioral economics [e.g., 20] – this dissertation will take an approach to framing that is in line with the perspective from sociological studies [e.g., 9, 16]. These studies treat framing as a dynamic process by which people interpret and construct their understanding of events around them.

designs for computational techniques to help identify the language associated with framing. It also will conduct user studies to evaluate how the language identified using the proposed models might assist researchers with conducting framing analysis. Next, it will utilize the proposed techniques to examine the ways false and misleading content might play a role in framing processes.

3.1 Investigating Computational Techniques to Facilitate Analyzing Framing Processes

Drawn from the definition of framing [16, 17] and motivated by prior computational work on identifying frame evoking attributes [e.g., 8, 32], this chapter of my dissertation will examine the relevance of different linguistic attributes, such as word choices, thematic patterns, and grammatical structures. It will also examine the efficacy of different models based on these attributes to identify language indicative of framing processes.

To investigate the effectiveness of the proposed attributes, this study begins with employing an unsupervised topic modeling technique, as utilized in prior research [11, 32]. Next, it extends this technique to incorporate additional linguistic attributes not integrated in prior models.

This work will conduct two studies to evaluate which of the proposed model might most effectively assist researchers with analyzing processes involved in framing. First, the models will be evaluated in a qualitative interview to understand how the patterns identified resemble the aspects researchers consider when analyzing processes involved in framing, and how these patterns might be used to facilitate the analysis of framing processes. Second, the models will be further evaluated in a quantitative study to understand how patterns identified using the proposed models can assist researchers in analyzing specific aspects of framing processes.

To reiterate, the computational techniques proposed in this chapter do not intend to identify framing processes directly. Rather, they are intended to identify text-oriented intuitions that might be indicative of framing and can assist researchers when examining the processes of framing.

3.2 Investigating the Involvement of Misinformation in Framing Processes

Using the computational techniques developed in the previous chapter, this chapter of my dissertation will examine the relationships between the processes of framing and the phenomenon of misinformation. Specifically, it will examine whether and how the way false and misleading content interplays with community responses might contribute to the wider changes in the way a community perceives and interprets the world's events and respond to the surrounding issues.

To do so, it will take an observational approach and will explore framing processes in two communities with varying levels of false and misleading content prevalence. Specifically, it will examine whether there are any meaningful differences in the ways online communities with varying degrees of false and misleading content might respond to the framing of an event depicted in the mainstream, in ways that reinforce, alter, or even entirely reject those framing processes.

To explore this question, this study will focus on the COVID-19 pandemic as a testbed. Indeed, the COVID-19 pandemic presents a unique opportunity to explore this question given the abundance of false and misleading content during this time [14, 25] and the impact such content had on how people view, interpret (i.e., frame) and respond to the pandemic and its various aspects.

The potential findings of this study contribute to expanding knowledge around the role of misinformation in the way individuals and online communities interpret the world's events, as well as their positions in relation to these events. These insights can inform the development of interventions aimed at mitigating such broad, community-level impacts of misinformation.

4 EXPECTED CONTRIBUTIONS

My dissertation will contribute to a more nuanced understanding of the complex phenomenon of misinformation and its broad impacts at the community level, beyond simply misleading individuals about individual pieces of false and misleading content. To do so, it adopts an ecological perspective to the phenomenon of misinformation, and account for the community oriented aspects of misinformation and its impacts. Specifically, it demonstrates how misinformation shapes perceptions about a community, thereby contributing to significant consequences at the community level. It also shows how community response to false and misleading content can play a significant role in mitigating such broad impacts.

In addition, my work will examine how false and misleading content could play a role in the processes of meaning constructions (i.e., framing) and impact the ways individuals and communities respond to the world's events. The potential findings of this dissertation can provide valuable insights into the design of interventions that address the broader impacts of misinformation, beyond addressing individual pieces of false and misleading content. In this way, my work will demonstrate how adopting an ecological perspective to the phenomenon of misinformation may enable us to attend to the broader scope of misinformation and its extensive impacts.

ACKNOWLEDGMENTS

I am grateful for the continuous guidance of my advisor, Eric P. S. Baumer (Lehigh University, Computer Science Department), and to my committee Dominic DiFranzo (Lehigh University, Computer Science Department), Su Lin Blodgett (Microsoft Research, FATE) and Brian D. Davison (Lehigh University, Computer Science Department). This material is based in part upon work supported by the National Science Foundation under Award No. 2106476. and Award No. 2212265.

REFERENCES

- [1] 2017. Fact-checking fake news on Facebook works - just too slowly. <https://apnews.com/article/north-america-technology-business-fake-news-ap-fact-check-e03283b4169f4d8c8a7e51042d61bcb5>.
- [2] 2018. How whatsapp helped turn an Indian village into a lynch mob. <https://www.bbc.com/news/world-asia-india-44856910>
- [3] 2020. Expanding Facebook's U.S. Fact-Checking Program and Supporting the Fact-Checking Ecosystem. <https://www.facebook.com/journalismproject/fact-checking-expansion-and-investment-2020>
- [4] 2020. Fake news and its impact on the economy. <https://priorityconsultants.com/blog/fake-news-and-its-impact-on-the-economy/>.
- [5] 2021. Twitter launches crowd-sourced fact-checking project. <https://apnews.com/article/twitter-launch-crowd-sourced-fact-check-589809d4c9a7eceda1ea8293b0a14af2>

- [6] Zhila Aghajari, Eric P. S. Baumer, and Dominic DiFranzo. 2023. Reviewing Interventions to Address Misinformation: The Need to Expand Our Vision Beyond an Individualistic Focus. (2023).
- [7] Zhila Aghajari, Eric P. S. Baumer, and Dominic DiFranzo. 2023. What's the Norm Around Here? Individuals' Responses Can Mitigate the Effects of Misinformation Prevalence in Shaping Perceptions of a Community. (2023).
- [8] Eric P. S. Baumer, Vera Khovanskaya, Mark Matthews, Lindsay Reynolds, Victoria Schwanda Sosik, and Geri Gay. 2014. Reviewing reflection: on the use of reflection in interactive system design. In *Proceedings of the 2014 conference on Designing interactive systems*. 93–102.
- [9] Robert D Benford and David A Snow. 2000. Framing processes and social movements: An overview and assessment. *Annual review of sociology* (2000), 611–639.
- [10] Md Momen Bhuiyan, Michael Horning, Sang Won Lee, and Tanushree Mitra. 2021. NudgeCred: Supporting News Credibility Assessment on Social Media Through Nudges. *Proceedings of the ACM on Human-Computer Interaction* 5, CSCW2 (2021), 1–30.
- [11] David M Blei, Andrew Y Ng, and Michael I Jordan. 2003. Latent dirichlet allocation. *Journal of machine Learning research* 3, Jan (2003), 993–1022.
- [12] Alexandre Bovet and Hernán A Makse. 2019. Influence of fake news in Twitter during the 2016 US presidential election. *Nature communications* 10, 1 (2019), 1–14.
- [13] Jonas Colliander. 2019. “This is fake news”: Investigating the role of conformity to other users' views when commenting on and spreading disinformation in social media. *Computers in Human Behavior* 97 (2019), 202–215.
- [14] Adam M Enders, Joseph E Uscinski, Casey Klofstad, and Justin Stoler. 2020. The different forms of COVID-19 misinformation and their consequences. *The Harvard Kennedy School Misinformation Review* (2020).
- [15] Robert M Entman. 1993. Framing: Towards clarification of a fractured paradigm. *McQuail's reader in mass communication theory* 390 (1993), 397.
- [16] William A Gamson. 1989. News as framing: Comments on Graber. *American behavioral scientist* 33, 2 (1989), 157–161.
- [17] William A Gamson and Andre Modigliani. 1989. Media discourse and public opinion on nuclear power: A constructionist approach. *American journal of sociology* 95, 1 (1989), 1–37.
- [18] Erving Goffman. 1974. *Frame analysis: An essay on the organization of experience*. Harvard University Press.
- [19] Andrew M Guess and Benjamin A Lyons. 2020. Misinformation, disinformation, and online propaganda. *Social media and democracy: The state of the field, prospects for reform* 10 (2020).
- [20] Daniel Kahneman and Amos Tversky. 1984. Choices, values, and frames. *American psychologist* 39, 4 (1984), 341.
- [21] Alex Zhi-Xiong Koo, Min-Hsin Su, SangWon Lee, So-Yun Ahn, and Hernando Rojas. 2021. What Motivates People to Correct Misinformation? Examining the Effects of Third-person Perceptions and Perceived Norms. *Journal of Broadcasting & Electronic Media* (2021), 1–24.
- [22] David MJ Lazer, Matthew A Baum, Yochai Benkler, Adam J Berinsky, Kelly M Greenhill, Filippo Menczer, Miriam J Metzger, Brendan Nyhan, Gordon Pennycook, David Rothschild, et al. 2018. The science of fake news. *Science* 359, 6380 (2018), 1094–1096.
- [23] Stephan Lewandowsky, Ullrich KH Ecker, and John Cook. 2017. Beyond misinformation: Understanding and coping with the “post-truth” era. *Journal of applied research in memory and cognition* 6, 4 (2017), 353–369.
- [24] Stephan Lewandowsky, Ullrich KH Ecker, Colleen M Seifert, Norbert Schwarz, and John Cook. 2012. Misinformation and its correction: Continued influence and successful debiasing. *Psychological science in the public interest* 13, 3 (2012), 106–131.
- [25] Sahil Loomba, Alexandre de Figueiredo, Simon J Piatek, Kristen de Graaf, and Heidi J Larson. 2021. Measuring the impact of COVID-19 vaccine misinformation on vaccination intent in the UK and USA. *Nature human behaviour* 5, 3 (2021), 337–348.
- [26] Gordon Pennycook and David G Rand. 2021. Examining false beliefs about voter fraud in the wake of the 2020 Presidential Election. *The Harvard Kennedy School Misinformation Review* (2021).
- [27] Shruti Phadke, Mattia Samory, and Tanushree Mitra. 2021. What Makes People Join Conspiracy Communities? Role of Social Factors in Conspiracy Engagement. *Proceedings of the ACM on Human-Computer Interaction* 4, CSCW3 (2021).
- [28] Ankita Rao. 2021. Guardian News and Media. (2021, January 5). US pharmacist who tried to ruin Covid vaccine doses is a conspiracy THEORIST, police say. <https://www.theguardian.com/us-news/2021/jan/04/wisconsin-pharmacist-covid-19-vaccine-doses-steven-brandenburg>.
- [29] Dietram A Scheufele. 2000. Agenda-setting, priming, and framing revisited: Another look at cognitive effects of political communication. *Mass communication & society* 3, 2-3 (2000), 297–316.
- [30] Briony Swire, Adam J Berinsky, Stephan Lewandowsky, and Ullrich KH Ecker. 2017. Processing political misinformation: comprehending the Trump phenomenon. *Royal Society open science* 4, 3 (2017), 160802.
- [31] Amos Tversky and Daniel Kahneman. 1985. *The framing of decisions and the psychology of choice*. Springer.
- [32] Dror Walter and Yotam Ophir. 2019. News frame analysis: An inductive mixed-method computational approach. *Communication Methods and Measures* 13, 4 (2019), 248–266.
- [33] Claire Wardle, Hossein Derakhshan, et al. 2018. Thinking about ‘information disorder’: formats of misinformation, disinformation, and mal-information. *Ireton, Cherilyn; Posetti, Julie. Journalism, fake news & disinformation. Paris: Unesco* (2018), 43–54.